

LINUX FILE PERMISSIONS



WHY FILE PERMISSIONS ?

LINUX....

- MULTITASKING SYSTEM
- MULTIUSER SYSTEM
- Permissions can be set to files and Directories.



chmod COMMAND

- *To change the permissions of a file or directory*
- *Syntax:*

chmod assignment_expression filename

- *Assignment expression comprises* 

- *Category of users*
- *Granting or denial of permission*
- *Types of users*



Types of Users

- Owner - owner of the file or directory.
- Group - group of related users.
- All/Other- Other users.



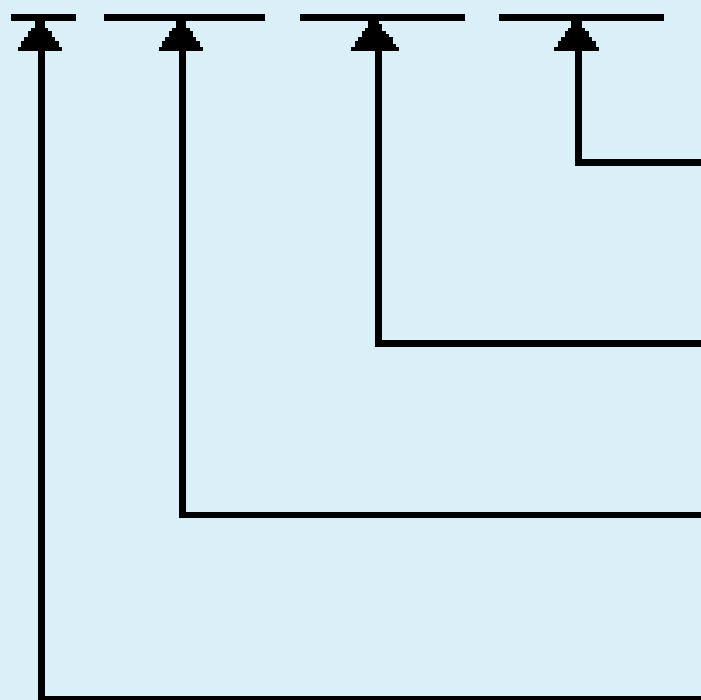
Types of Permissions

- Read - read the file or list files in the directory
- Write - add new content to a file or add new files to the directory
- Execute - access the contents of a file or access files in the directory



WHILE LISTING....

- rwxrw - r - -



Read, write, and execute permissions
for all other users

Read, write and execute permissions
for members of the group owning the
file.

Read, write and execute permissions
for the owner of the file.

File type. "-" indicates a regular file. A
"d" indicates a directory.

PERMISSION VALUES

r	w	x	Octal Value	Meaning
0	0	0	0	No permission
0	0	1	1	Execute-only permission
0	1	0	2	Write-only permission
0	1	1	3	Write and execute permissions
1	0	0	4	Read-only permission
1	0	1	5	Read and execute permissions
1	1	0	6	Read and write permissions
1	1	1	7	Read, write, and execute permissions

VALUES FOR SYMBOLIC MODES

Who	Operator	Privilege
u User	+ Add privilege	r Read bit
g Group	- Remove privilege	w Write bit
o Other	= Set privilege	x Execute/search bit
a All		u User's current privileges
ugo All		g Group's current privileges
		o Others' current privileges
		l Locking privilege bit
		s Sets user or group ID mode bit
		t Sticky bit

Chmod

- SINGLE USER
- Example :
- Cmd: `chmod u + r filename`
- The above command grants permission to a single user to read the file .



Chmod

MULTIUSERS

- *Example :*
- *Cmd : chmod u + rw , g = x , o -w filename*
- *The above command grants permissions to multiple users.*



OCTAL NOTATION

- ✓ *Cmd: chmod 761 filename*
- ✓ *Cmd: chmod u = rwx, g = rw, o = x filename*
- ✓ *The above commands are equivalent to each other.*



EXAMPLES USING OCTAL VALUES

Command	Purpose
<code>chmod 700 *</code>	Sets access privileges for all the files (including directories) in the current directory to read, write, and execute for the owner, and provides no access privilege to anyone else
<code>chmod 740 courses</code>	Sets access privileges for courses to read, write, and execute for the owner and read-only for the group, and provides no access for others
<code>chmod 751 ~/courses</code>	Sets access privileges for ~/courses to read, write, and execute for the owner, read and search for the group, and search-only permission for others
<code>chmod 700 ~</code>	Sets access privileges for the home directory to read, write, and execute for the owner, and no privileges for anyone else. (The authors use this command for their home directories; they do not trust anyone when it comes to security of their files, and they stop everyone at the main gate.)

EXAMPLES USING SYMBOLIC NOTATIONS

<code>chmod u=rwx courses</code>	Sets owner's access privileges to read, write, and execute for courses and keeps the group's and others' privileges to their present values
<code>chmod ugo-rw sample</code> or <code>chmod a-rw sample</code>	Does not let anyone read or write sample
<code>chmod a+x sample</code>	Lets everyone execute sample
<code>chmod g=u sample</code>	Makes sample's group privileges match its user (owner) privileges
<code>chmod go= sample</code>	Removes all access privileges for the group and others for sample

Chown command

- To change the owner / user of the file / directory.
- An administrator command
- Only the root user can change the owner of the file / directory.



- Syntax:

chown [options] newowner file/dir name



Chgrp command

- To change the group of a file / directory.
- An administrator command.
- Syntax:

chgrp [options] newgroup file/dirname



THANK YOU

